

Lab 3: Architectural Justification

Hospital Bed & ICU Allocation Dashboard
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1. Architecture Style Comparison

Architecture	Advantages for this scenario	Limitations / fit
Layered	Clear separation of presentation, business logic and data; straightforward structure.	Some redundancy; chosen for the centralized dashboard.
Microservices	Independent scaling and fault isolation.	Operational complexity and network/data-consistency overhead can be excessive.
Client-Server	Centralized control and simple deployment.	Server can become a single point of failure and scalability bottleneck.

2. Architecture Selection

We chose Layered Architecture for the Hospital Bed & ICU Allocation Dashboard. The system is organized into a Presentation Layer, Business Logic Layer and Data Layer, with the component interactions shown in the UML component diagram.

3. Reason 1 — Clear separation of responsibilities

The dashboard has distinct responsibilities for user interaction, triage and patient prioritization, bed/ICU management, transfer and notification handling, and data storage. Layering keeps these responsibilities separated and makes the system easier to understand, maintain and modify.

4. Reason 2 — Suitable for the centralized dashboard

Hospital staff use one dashboard for bed allocation, occupancy monitoring and transfer coordination. A layered structure provides a direct flow from the user interface through business logic to persistent hospital data, matching the project developed in Labs 1 and 2.

5. Security Advantage

Security controls can be enforced at the application/business layer so restricted triage, bed-allocation and transfer operations are available only to authenticated and authorized users. Database access is kept behind the business layer.

6. Performance Benefit

The separated layers allow dashboard operations such as bed-status updates and occupancy retrieval to follow a direct application path. This supports the Lab 1 requirement that bed-status transitions reach connected dashboard clients in under 1 second via WebSockets.

Diagram check: exactly 5 components and exactly 4 interfaces are shown in the accompanying UML component diagram.